# Modeling distance in pods

*Authors:* Averey Ku and Siddha Karjee

## Generative AI (genAI) Usage Statement

Write a short statement about how you used genAI on this assignment. *Note*: If you don't write anything, we'll assume you used genAI to write the whole thing!!!

*(Write your response here)*

We did not use generative AI on this assignment.

## 1. Modeling Question

*(State your modeling question directly. Categorize it according to the action/specificity framework.)*

*(You will likely need to write 2-3 paragraphs for this section.)*

*Suggested template: FOR INSPIRATION, NOT TO USE DIRECTLY.*

*A: we were curious about some question because of some reasons*

*B: but we weren't sure how to model it*

*T: so we did some background research for ideas (i.e. research synthesis matrix)*

*While we were conducting background research for our previous modeling question about the length of an infection wave in Olin versus Needham, we learned about various mathematical models for infectivity as a function of distance for individual-to-individual transmission. However, we didn't find as much information about large-scale non-homogeneous populations, which are a more accurate representation of reality. Thus we decided to tackle the question of whether there is a mathematical relationship between the peak number of infected individuals as a function of distance and individual-to-individual infectivity as a function of distance. In other words, given individual-to-individual infectivity as a function of distance, can we predict the peak number of infected individuals for the same distance value?*

*On the specificity-action framework, we believe this model to be relatively general, since it is applicable to any non-homogeneous population, and rhetorical, since we cannot directly use the results from our model to do anything.*

## 2. Literature Search

*After looking through more articles, including some different approaches to modeling non-homogeneous populations, we decided to summarize our results in a research synthesis matrix.*

## 3. Natural World

*In the real-world, infection rate, mortality rate, and recovery rates are complex non-linear functions of many factors including individual character, social environment, infection route, and intrinsic properties of the virus [1]. Also, real-world populations are also non-homogeneous, meaning the strength of connection between individuals is not uniform [2]. For example, cities have very different social connectivity when compared to towns or rural areas. Finally, many viruses in the real-world can be fatal, such as ebola, rabies, and smallpox.*

*However, we assumed these to be irrelevant for our modeling question and so we left them outside the system boundary.*

## 4. Model: Representation & Assumptions

*For our model, however, we assumed that infection rate is an inverse exponential function of distance multiplied by some fixed rate representing the infectivity of the virus. We believe this to be a valid simplifying assumption because distance is a major factor in infectivity rate, with Sun et al. finding that halving occupational density in indoor spaces reduced infection rate by 20-40% [3]. We also decided to represent the real-world with a uniformly podded network rather than a more nuanced social connectivity network as a part of our modeling question for feasibility reasons, getting .*

*Undirected vertex-edge graphs are the basis of our model. There are different kinds of these graphs, including fully connected (meaning that every node is directly or indirectly connected to every other node) and disconnected graphs (where at least vertices exist with no path connecting them).*

*Network graphs are a specific version of vertex-edge graphs where nodes represent individuals (or households) and edges represent connections between these nodes. By assigning a weight to each edge, stronger connections between certain individuals can be represented (such as those in the same household.)*

*There are multiple ways to represent un-directed graphs including edge lists, adjacency lists, or adjacency matrices. Edge lists are the most straightforward way, storing all edges as a list of pairs of endpoint 1 and endpoint 2. Adjacency lists store a collection of neighbors for each vertex using a dictionary, hash map, or an array of lists [4]. This is the most common way of representing graphs and is used for most graph search algorithms such as DFS and BFS because of its memory efficiency. Finally, adjacency matrices store the strength of connection between two nodes u and v at the uth row and vth column. It is the fastest method since retrieving edges takes constant time (matrix indexing is O(1)). It however also has a high memory cost taking up O(n^2) regardless of how many edges exist, making it best suited for small and dense graphs.*

*We chose to represent our graph with an adjacency matrix because it is relatively small and dense (each pod is connected to every other pod).*

*Our parameters for the model are num_trials, t_end, min_distance, max_distance, infection_rate, recovery_rate and num_steps. We decided to add a num_trials parameter because the agent-based SIR model is probabilistic and averaging out the output number of infected individuals I_h produces more stable results. We heuristically chose a value of 20 for num_trials in order to balance between computational resources (and more importantly runtime :)) and stability of the outputs. The second parameter for our model is t_end, which represents the time frame for simulation. We chose ____ for t_end in order to fully simulate the infection wave for all of the different values of distance between pods, which we parameter-sweeped from a min_distance of 50 to a max distance of 200 with 50 steps. Finally we arbitrarily chose an infection rate and recovery rate so that we could clearly see the different between the different curves of infected individuals for each distance.*

```
% affects transmission for the distance between pods
function strength = distancetostrength(d)
strength = exp(-d/50);

end

function M_res = make_adj_pods(n_per_pod, n_pods, dBlock)
% Construct an adjacency matrix for a "podding" scenario
%
% Usage
%   M_res = make_adj_pods(n_per_pod, n_pods, infectivity)
%
% Arguments
%   n_per_pod  = number of persons per "pod"
%   n_pods     = total number of pods; total population = n_per_pod * n_pods
%   pod_effect = relative infectivity within pods; should be infectivity > 1
%
% Returns
%   M_res = adjacency matrix for desired graph
n_total = n_per_pod * n_pods;

% Create the "base matrix" with weak connections between all persons
M_res =  distancetostrength(dBlock)* (ones(n_total, n_total) - eye(n_total));

% Add the pod structure
```

```
for i = 1:n_pods
    % Determine the slicing indices
    i0 = (i - 1) * n_per_pod + 1; ie = i0 + n_per_pod - 1;
    % Assign strong in-pod weights
    M_res(i0:ie, i0:ie) = 10 * M_res(i0:ie, i0:ie);
end

end



initial_state = zeros(120, 1);
initial_state(1:10) = 1;



t_end = 120;
infection_rate = 0.005;
recovery_rate = 0.05;

% number of different distances
num_steps = 50;

% range of possible distances
min_distance = 50; max_distance = 200;
num_trials = 20;

% array of possible distances
distance_vals = linspace(min_distance, max_distance, num_steps);
I_mean = zeros(num_steps, t_end);


%example matrix network graph
ex_matrix = make_adj_pods(20, 6, 100);
Graph_6pod = graph(ex_matrix);
figure();
h = plot(Graph_6pod, 'NodeColor', 'k', 'LineWidth', 0.1);
```

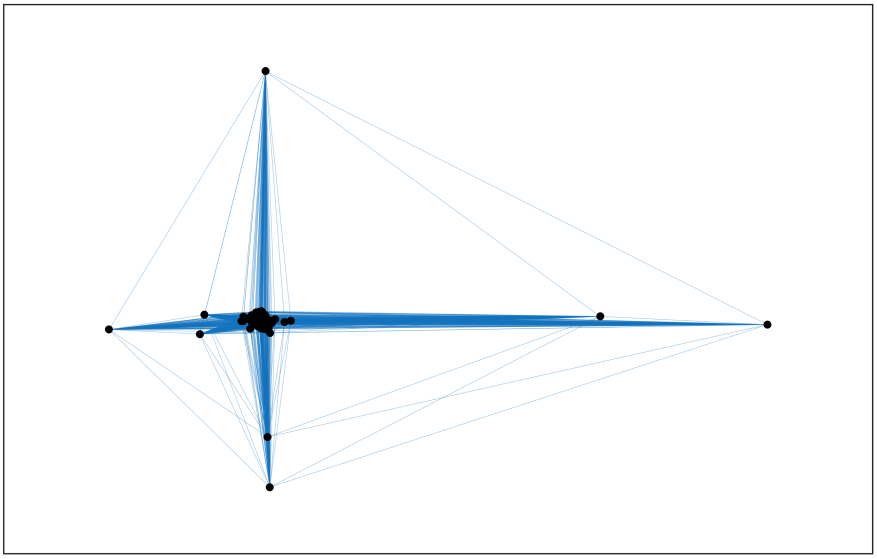


***Figure 1.*** *Visualization of example graph network for distance of 100 feet*

*Finally to actually model an infection spreading through our network, we used an agent-based SIR model. This is a variation of the classic compartmental model, where the state of each individual is tracked in a vector. In order to optimize performance, the time series of infected vectors is horizontally stacked into a matrix.*

```
function [Sh, Ih, Rh] = simulate_absir(M, Iv0, T, infection_rate, recovery_rate)
% Simulate agent-based transmission model. Uses a graph to represent social
% connectivity between agents.
%
% Note: You may find it helpful to summarize the state history matrices via
% summation. For instance sum(Ih, 1) will return the total number of
% infected persons at each timestep in the simulation.
%
% Inputs
%   M (square matrix): Adjacency matrix
%   Iv0 (column vector): Initial infection state
%   T (integer): Number of timesteps to simulate
%   infection_rate (float): Infection rate (probabilistic)
%   recovery_rate (float): Recovery rate (probabilistic)
%
% Returns
%   Sh (matrix): Susceptible state history
%   Ih (matrix): Infected state history
%   Rh (matrix): Recovered state history

% Setup
dim = length(Iv0);  % Dimensions of initial state
Ih = zeros(dim, T); % Infection history
```

```
Ih(:, 1) = Iv0; % Record the initial state to infection history
Rh = zeros(dim, T); % Recovery history

% Construct an "action" helper function
    function [I, R] = action(I, R)
        % Compute infection probabilities based on the social graph
        v_eff = infection_rate * M * I;
        v_pr = v_eff ./ (1 + v_eff);

        % Draw random values
        v_infect = rand(dim, 1) <= v_pr;
        v_recover = rand(dim, 1) <= recovery_rate;

        % Infect non-recovered individuals
        I = I | v_infect & (~R);
        % Recover infected individuals
        R = R | (I & v_recover);
        I = I & (~R);
    end

% Run simulation
for i = 2:T
    [Ih(:, i), Rh(:, i)] = action(Ih(:, i-1), Rh(:, i-1));
end

% Compute susceptible history
Sh = ones(dim, T) - Ih - Rh;
end
```

*In conclusion we decided to implement our model as a edge-weighted graph with 6 pods of 20 nodes each and the weight calculated as an inverse exponential of the distance between individuals multiplied by a fixed infectivity rate. We then ran an SIR model for 120 timestep for each value of distance ranging from 50 to 200 feet at intervals of 3 feet.*

```
%simulates every distance from 50-200
for k = 1:num_steps
    adj_matrix = make_adj_pods(20, 6, distance_vals(k));
    for p = 1:num_trials
        [~, Ih, ~] = simulate_absir(adj_matrix, initial_state, t_end,
infection_rate, recovery_rate);
        I_mean(k,:) = I_mean(k,:) + sum(Ih, 1) / num_trials;
    end
end
```

## 5. Verification & Validation

*(Select appropriate verification facts, and implement them as assert statements. In Projects 2 and 3, carry out model validation.)*

*(You will likely need to write 2-3 code sections for this section, and use words to describe your argument for verification and validation.)*

*Suggested template: FOR INSPIRATION, NOT TO USE DIRECTLY.*

*One fact that should be true is that the number of infected people doesn't exceed the population, and also, we need to make sure that the distance coefficient decreases the transmission of the virus.*

```
%makes sure Infected count doesn't exceeds population
assert(all(sum(Ih, 1) <= 120))
%makes sure that the distance decreases the transmission of the matrix
assert(distancetostrength(min_distance) > distancetostrength(max_distance))
```

```
% plots every 5th graph
I_plot = I_mean(1:5:end, :);
figure
plot(1:t_end, I_plot)
xlabel('Timestep (in weeks)'); ylabel('Infected (in people)')
```

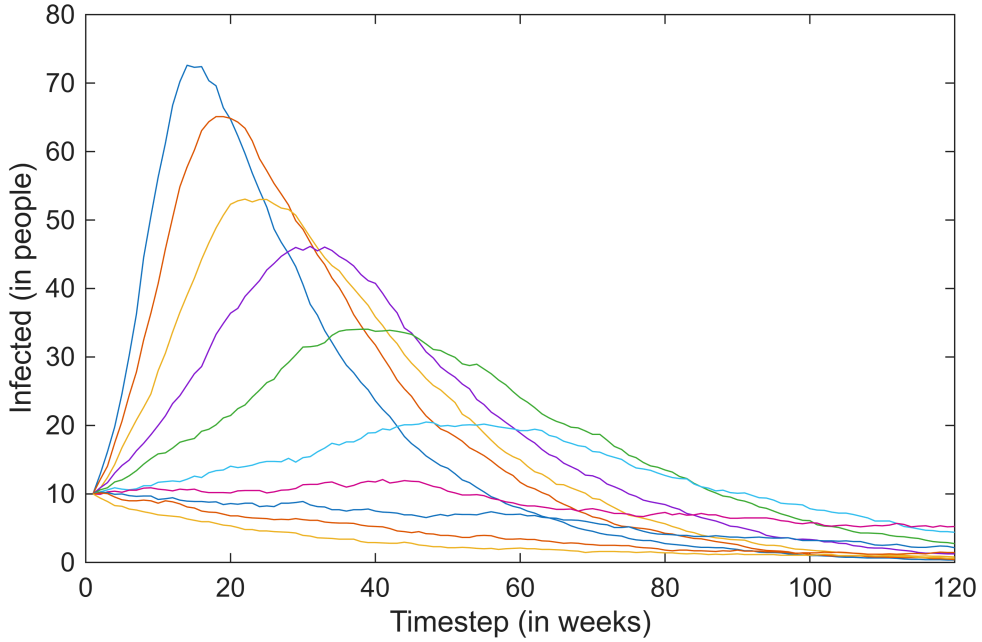



***Figure 2**. Simulated results of infected individual count over time with different distances (blue is 50 feet and orange is 200 feet).*

*Note: every fifth curve was plotted in order to preserve legibility.*

***Peak Infected***

```
peaks = max(I_mean, [], 2);

[p,coeffs] = polyfit(distance_vals, peaks, 4);

f = polyval(p, distance_vals);
figure
plot(distance_vals, peaks)
hold on
plot(distance_vals, f)
xlabel('Distance (in feet)'); ylabel('Peak Infected (in people)')
```

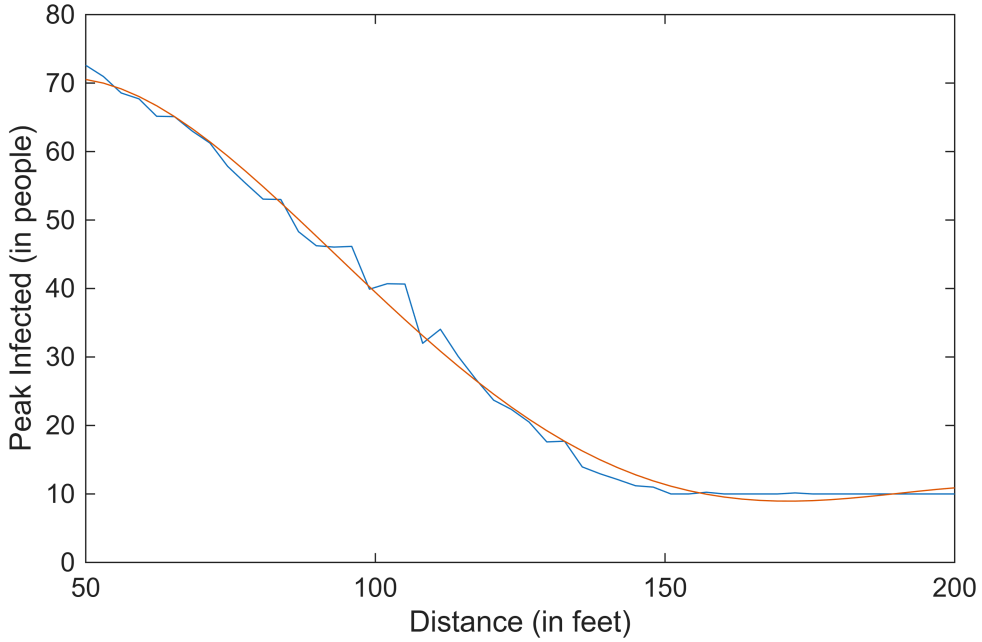



***Figure 3**. Peak infected individual count plotted against each value of distance with a exponential fit*

```
% calculates the r^2 value
R_squared = 1-(coeffs.normr/norm(peaks - mean(peaks)))^2
```

```
R_squared =
0.9952
```

## 6. Claims & Limitations

*One assumption from Section 4 was that infection rate is only a function of distance and intrinsic infectivity rate. Despite this assumption however, we found that our peak infected as a function of distance was best fit by a polynomial (R^2 of .992). So we can conclude that there was no obvious relationship between our infection strength as a function of distance, and our peak infected as a function of distance, keeping in mind that this result may not be true in cases where the infection route or the social environment is different.*

*Another assumption from Section 4 that population evenly podded (i.e pods of equal size and equal connections between and within the pods.) With our results listed above, we need to keep in mind that we won't be able to model different kinds of populations, e.g in cities.*

## 7. Conclusion

*Our modeling question was that given individual-to-individual infectivity as a function of distance, can we predict the peak number of infected individuals for the same distance value, but we were able to determine that polynomial fit gave us the best fit for the graph of peak infected vs distance, whereas for strength, we used an exponential function for distance. Additionally, the polynomial function with degree four gave us a r^2 value of 0.9934, which is why we determined that the polynomial graph would be the best fit. In conclusion, there is no mathematical relation between infection rate as a function of distance and peak infections in pods a function of distance.*

## 8. Future Work

*Our model assumed that infection rate is only a function of distance and intrinsic infectivity rate. But in the natural world, there are a lot more factors, including individual character, social environment, infection route, and intrinsic properties of the virus. Thus an interesting follow-up question would be how can infection rate be written as a function of individual character, social environment, infection route, and intrinsic properties of the virus?*

*Our model also assumed that the population could be represented through 6 pods of 20 individuals. However in the natural world, the population network graph is much more complex, with connections being stronger in more densely population areas and vice versa. An interesting follow-up would be the same question, implemented with a different adjacency matrix based on real-world data from a city.*

## References

*(1) Manriquez R. et al, Spread of Epidemic Disease on Edge-Weighted Graphs from a Database: A Case Study of COVID-19*

*(2) Park S. et al, Infection dynamics for fluctuating infection or removal rates regarding the number of infected and susceptible individuals*

*(3) Sun C. et al, The efficacy of social distance and ventilation effectiveness in preventing COVID-19 transmission*

*(4) Adjacency List Representation, https://www.geeksforgeeks.org/dsa/adjacency-list-meaning-definition-in-dsa/*